Reveal Minesweeper

AlgoExpert · Medium · Recursion

The problem

- Board of "M" (mine), "H" (hidden), "0"-"8" (revealed).
- Clicking a mine sets it to "X"; clicking a hidden cell reveals its adjacent-mine count and, if zero, recursively reveals neighbours.

Intuition

- Clicking a mine ends the game; clicking an empty square reveals neighbours recursively.

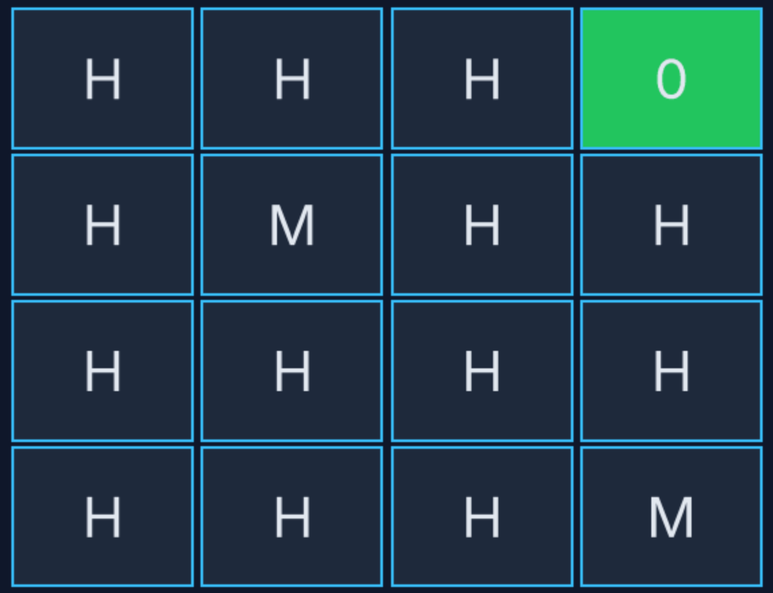
Brute force vs optimal

- Click a mine → X and stop
- Count neighbours; if 0, flood-fill recursively
- Each cell revealed once → O(w·h)

Approach

- 1. If a mine, mark X
- 2. Count adjacent mines
- 3. If zero, reveal neighbours recursively
- 4. Otherwise show the count

Worked example



Reveal (0,3) → 0

Worked example



Reveal (0,2) → 1

Worked example



Reveal (1,2) → 1

Worked example



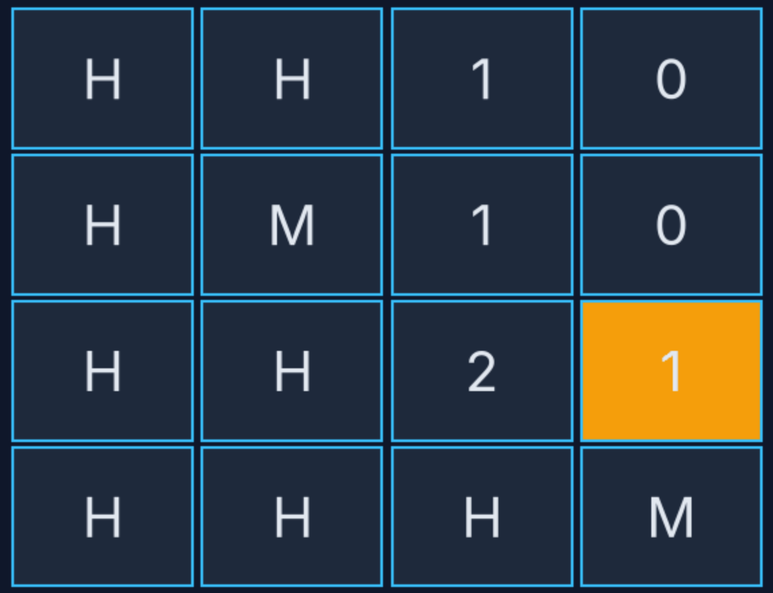
Reveal (1,3) → 0

Worked example



Reveal (2,2) → 2

Worked example



Reveal (2,3) → 1

Java solution

```java
public class RevealMinesweeper {
    public static String[][] revealMinesweeper(String[][] board, int row, int col) {
        if (board[row][col].equals("M")) { board[row][col] = "X"; return board; }
        reveal(board, row, col);
        return board;
    }

    private static void reveal(String[][] b, int r, int c) {
        if (!b[r][c].equals("H")) return;
        int mines = 0;
        for (int dr = -1; dr <= 1; dr++) for (int dc = -1; dc <= 1; dc++) {
            int nr = r + dr, nc = c + dc;
            if ((dr != 0 || dc != 0) && nr >= 0 && nc >= 0 && nr < b.length && nc < b[0].length && b[nr][nc].equals("M")) mines++;
        }
        b[r][c] = String.valueOf(mines);
        if (mines > 0) return;
        for (int dr = -1; dr <= 1; dr++) for (int dc = -1; dc <= 1; dc++) {
            int nr = r + dr, nc = c + dc;
            if (nr >= 0 && nc >= 0 && nr < b.length && nc < b[0].length) reveal(b, nr, nc);
        }
    }
}
```

Complexity

- Time: O(w*h)
- Space: O(w*h)

Edge cases & pitfalls

- 8-directional neighbours
- Only recurse on hidden cells (no re-visits)
- Only recurse from zeros

Interview tips & follow-ups

- Mention BFS to avoid deep recursion
- Follow-up: generate a board, first click safe

Recap

- Clicking a mine ends the game; clicking an empty square reveals neighbours recursively.
- Complexity: O(w*h)